

Ruiqing Tang

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Education

Master in Communication Engineering	2023.09 – 2026.06
University of Science and Technology, Beijing	Beijing, China
Bachelor in Communication Engineering	2019.09 - 2023.06
University of Science and Technology, Beijing	Beijing, China

Work Experience

3D AIGC Algorithm Engineer	2026.07 – present
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[Yuanjing, Alibaba Group](#), Shanghai, China

● 3D Infinite Canvas — Agentic 3D Content-Production Platform

- Taking text/image references to game-ready 3D assets: an agent autonomously operates infinite-canvas generation, baking and scene assembly through typed tools and hard human-review gates.
- Architected the agent platform on an in-process multi-provider LLM core: a typed tool registry bridging 80+ canvas/3D/bake operations, with per-tool manifests (execution location, mutating vs. read-only, timeout, parallel-safety); migrated 17 long-running 3D/Hunyuan/baking tools to server-side execution so submitted jobs survive browser close and reconnect via original jobId; delivered protocol-v1 SSE streaming with reasoning/response separation.
 - Hardened long-running generation workflows: provider+jobId-keyed task recovery across browser close, network interruption and retry (zero duplicate paid resubmission); concurrent submission deduplication by content digest; revision/requestId conflict checks on scene writes so the agent cannot overwrite user-locked or stale state; skill-scoped prompt injection with hard human-review gates.
 - Built the agent's 3D pipeline tools: 3D generation across Hunyuan3D/Trellis2/Pixel3D providers with target-face control (default 500K faces); a six-stage baking pipeline (decimation retopology component fix UV smoothing texture bake) as composable canvas nodes, each independently re-runnable and resume-safe by taskId; multi-provider 3D job recovery keyed on provider+jobId+targetFaces.

3D AIGC Algorithm Internship	2026.03 – 2026.06
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[Yuanjing, Alibaba Group](#), Beijing, China

- Achieved an Agentic 3D AIGC workflow automating text/image to game-ready model conversion. Engineered autonomous requirement decomposition and multi-tool (decimation, retopology, UV unwrapping, smoothing, bake, format conversion) orchestration to enable fully automated 3D asset production from natural language prompts and images. Reduced meshes from ~500K to 3K–20K faces (25–160× compression) while preserving visual fidelity via high-to-low texture baking; delivered GLB/FBX outputs compatible with Unreal and Unity import pipelines.

3D Digital Human Generation Algorithm Intern

2025.07 – 2025.12

[AiShiWeiLai AI Research](#), Beijing, China

- Human Image Animation: Enhanced [Stable Video Diffusion](#) for robust cross-identity motion transfer.
 - Leveraged [GVHMR](#) and [HaMer](#) to extract 3D pose sequences; decoupled subject identity from SMPL parameters to derive universal skeletons, and injected reference skeleton IDs via cross-attention for identity-agnostic driving;
 - Dynamically fused [ArcFace](#) and [LivePortrait](#) embeddings with reference images through Cross Multi-Head Attention to inject facial and global context, significantly improving face-ID consistency and background stability.

AI Engineer Internship

2024.05 - 2025.06

[CubeVi](#), Beijing, China ([Product](#) & [Promo Video](#))

- 3D Reconstruction: Researched & reproduced SOTA 3D reconstruction techniques ([3DGS](#), [InstantSplat](#), etc.); Built production-ready meshing and stereo-photo generation tools for real scenes (buildings, natural landscapes, etc.), objects and human subjects, delivering immersive AR experience in the RealPlay app of the [Companion1](#).
- 3D Canvas: Developed an interactive 3D canvas enabling sketch-to-rigged-model generation: converts freehand sketches into 3D meshes with automatic skeletal rigging, and supports natural language-driven 3D animation creation.
- 3D Digital Human (Virtual Companion):
 - Motion Capture: Researched & reproduced SOTA 3D human motion capture algorithms ([SMPLer-X](#), [GVHMR](#), [HaMeR](#), etc.) and human motion generation algorithms ([MoMask](#), [LODGE](#), etc);
 - Animation Conversion: Developed end-to-end animation data conversion toolchains (SMPL $\Rightarrow$ FBX/VMD/BVH) with cross-engine compatibility (Babylon, Unity, Blender, Unreal

Engine), with SMPL2VMD latency reduced to just 400 ms for long-sequence animations (over 3 minutes).

- Audio: Researched and reproduced SOTA Text-to-Speech (TTS) techniques ([Chat-TTS](#), [GPT-SoVITS](#), etc.) and Singing Voice Conversion algorithms (e.g., [RVC](#), [seed-VC](#)); Crawled audio data for various characters, retrained models for each character, and optimized inference speed by converting models to ONNX format; GPT-SoVITS achieved ~1:6 latency (1s synthesis for 6s audio), while RVC voice separation and timbre conversion completed in under 1 minute.
- LLM&NPC: Developed a digital desktop pet application, integrating TTS, SVC, T2M, and LLM Role Play; Researched AI Agents for social NPC applications, implementing multi-user chatrooms with probability/rule-based behavior control; This digital desktop pet product has been launched in the [AI Life app of Companion1](#).

Projects and Research

3D Playmate

2025.02 – 2025.09

Personal Projects, Pursued Out of Interest.

- Conducted secondary development on LODGE, converting SMPL parameters into VMD animations and rendering them using the Three.js framework; Designed structured prompts (character profile, backstory, dialogue, skill attributes, and response constraints) to enable immersive Role Play; Demo presentation on different platforms: [youtube](#), [rednote](#), [bilibili](#).
- Proposed a shared MoE architecture for 3D dance generation, leveraging audio and dance-style embeddings to condition expert routing and enhance motion diversity and quality; Enforced Forward Kinematics constraints on foot joints to mitigate foot-skating artifacts. Further integrated PHC physics simulation to track generated motions in a physically plausible environment, significantly reducing penetration and joint distortion. Manuscript submitted to *AAAI 2027*.

SRTP (Student Research Training Program)

2020.12 - 2021.12

- Studied channel modeling for UAVs and explored communication protocols between UAVs, base stations, and ground terminals;
- Acquired foundational theoretical knowledge of reinforcement learning and applied the DPPO algorithm to optimize power allocation and trajectory planning for UAVs.

Academic & Reward

- [1] **R. Tang**, Y. Shi, F. Liu, X. Huang, I. Kamanga, H. Lv, T. Zhou, G. Tan, G. Zhu, H. Zeng, Y. Li, and X. Zhou, "A Lightweight Neural Network for Pipeline Flow Rate Measurement Using Distributed

- Acoustic Sensing," *IEEE Transactions on Instrumentation and Measurement (IEEE TIM)*, 2026, 75: 1-13.
- [2] I. Kamanga, **R. Tang**, Z. Wang, G. Zhu, F. Liu, and X. Zhou, "Performance enhancement of Φ -OTDR event classification via dynamic MFCCs and multi-scale discriminator-based FastGAN data augmentation," *Measurement*, 2026, 257(PE): 118927-118927.
- [3] H. Lv, F. Liu, **R. Tang**, I. Kamanga, G. Zhu, Y. Chen, A. Zhang, X. Yang, and X. Zhou, "Towards Open-Set Fault Diagnosis for Belt Conveyors Using Distributed Acoustic Sensing," *IEEE Sensors Journal*, 2026, 26: 23956-23969.
- [4] H. Lv, **R. Tang**, T. Zhou, Y. Shi, F. Liu, X. Huang, G. Zhu, I. Kamanga, and X. Zhou, "High-precision DAS vibration event recognition via a coupled wavelet transform-CNN framework," *Opt. Express*, 2026, 34: 32662-32676.
- [5] M. K. Shuvo, X. Huang, F. Liu, G. Zhu, **R. Tang**, K. Zhang, and X. Zhou, "Automatic hydraulic fracture hit events recognition with low-frequency DAS data based on object detection," *Optical Fiber Technology*, 2026, 103: 104780-104780.
- [6] **R. Tang**, X. Huang, F. Liu, H. Lv, Y. Wang, D. Li, T. Zhou, and X. Zhou, "Pipeline Flow Rate Monitoring Using Distributed Acoustic Sensing and Lightweight Neural Network," *Pacific Rim Conference on Lasers and Electro-Optics (CLEO-PR)*, 2026.
- [7] F. Liu, **R. Tang**, G. Tan, and X. Zhou, "A DAS pipeline flow monitoring method based on lightweight CNN," *Chinese Patent*, CN120043592A. 2025: 16. ([Published](#)).
- [8] H. Zhang, Y. Li, S. Tang, **R. Tang**, K. Long, P. Gao, and F. Li, "A method and device for allocating transmission power in an unmanned aerial vehicle network based on reinforcement learning," *Chinese Patent*, CN1-13316239B. 2022: 13. ([Granted](#)).
- [9] **R. Tang**, B. Hu, H. Li, Y. Liu, S. Zhang, W. Zhu, P. Cao, F. Liu, and X. Liu, "AniMorph: Cross-Identity and High-Fidelity Human Animation via Pose and Face Integration," *Neural Networks*, 2026, under review (Minor Revision).
- [10] Y. Chen, **R. Tang**, and Y. W., "PMDG: Physically-plausible Multi-genre Dance Generation via Diffusion and Robust Simulation," Association for the Advancement of Artificial Intelligence (AAAI), 2027, under review.
- [Second Prize](#), 4th Huawei Cloud Wireless Big Data Competition.

Skills

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- Languages: Python, C++, Shell.
 - Framework and libraries: Pytorch, Tensorflow, Scikit-learn, OpenCV, XGBoost, Blender, Unity, UE.